



User's Manual

*Web-Based 3D Printing Service Ordering Management System
For TechWorkz*

Online Education Department

October, 2025

**Abdulaziz N. Fitwi
Samuel C. Ogu**

TABLE OF CONTENTS

	<u>Page#</u>
1.0 GENERAL INFORMATION.....	1-1
1.1 System Overview.....	1-1
1.2 Project References.....	1-1
1.3 Authorized Use Permission.....	1-1
1.4 Points of Contact.....	1-1
1.5 Acronyms and Abbreviations.....	1-1
2.0 SYSTEM SUMMARY.....	2-1
2.1 Hardware and Software Specifications.....	2-1
2.2 User Access Levels.....	2-1
2.3 Contingencies and Alternate Modes of Operation.....	2-1
3.0 GETTING STARTED.....	3-1
3.1 Logging On.....	3-1
3.2 System Menu.....	3-1
3.3 Changing User ID and Password.....	3-1
3.4 Exit System.....	3-1
4.0 USING THE SYSTEM.....	4-1
4.1 Dashboard.....	4-1
4.2 Gallery.....	4-1
4.3 Ready-to-Print Model.....	4-1
4.4 Custom Order Submission.....	4-1
4.5 Cost Calculator.....	4-1
4.6 Order Tracking.....	4-1
4.7 Message/Chat.....	4-1
4.8 Payment & Verification.....	4-1
4.9 Report.....	4-1
4.10 Special Instructions for Error Correction.....	4-1
4.11 Cavats and Exception.....	4-1
5.0 QUERYING.....	5-1
5.1 Query Capabilities.....	5-1
5.2 Query Procedures.....	5-1
6.0 REPORTING.....	6-1
6.1 Generating Reports.....	6-1

1.0 GENERAL INFORMATION

1.0 GENERAL INFORMATION

1.1 System Overview

*The **Web-Based 3D Printing Service Ordering Management System** is an online platform designed to streamline customer order processing, enhance communication, and centralize record-keeping for 3D printing services. The system enables clients to:*

- *Register for an account or browse as a guest.*
- *View ready-to-print 3D model catalogs and previous projects.*
- *Upload custom 3D model files for printing.*
- *Automatically calculate order costs using a built-in 3D Printing Calculator.*
- *Track the real-time status of orders.*
- *Communicate with staff via an integrated chat module.*
- *Receive automated email notifications about order updates and payments.*

System Architecture:

The system is web-based and hosted on a cloud server, accessible through any standard web browser. It follows a client-server architecture, where users interact with a graphical user interface (GUI) while the backend processes handle order management, file storage, and reporting.

User Access Mode:

- *Clients/Guests: Web browser access (desktop/laptop recommended).*
- *Staff/Admin: Secure login with role-based access controls.*

Responsible Organization:

TechWorkz 3D Printing Services – Sunnycrest Village, Salitran II, Dasmariñas City, Cavite, Philippines.

System Environment & Special Conditions:

- *Requires a stable internet connection (minimum 10 Mbps or above).*
- *Best viewed on modern browsers (Chrome, Edge, Firefox, Safari).*
- *Designed for desktop use, but accessible on mobile devices.*

1.2 Project References

- Ogu, S.C., & Fitwi, A.N. (2025). *Web-Based 3D Printing Service Ordering Management System for TechWork bz – Capstone Project*, AMA University Online Education.

1.3 Authorized Use Permission

*This system is intended solely for the use of **TechWorkz 3D Printing Services**, its staff, and registered clients. Unauthorized access, duplication, or distribution of the system, its source code, data, or reports is strictly prohibited. Any attempt to modify or exploit the system without proper authorization will be subject to legal action.*

1.4 Points of Contact

Name: Samuel C. Ogu

Name: Abdulaziz N. Fitwi

Cellphone No.: 09495214834

Cellphone No.: 09542265968

Email: samuelogu01@gmail.com

1.5 Acronyms and Abbreviations

Acronym Meaning

3D - Three-Dimensional

AM - Additive Manufacturing

CAD - Computer-Aided Design

DBMS - Database Management System

GUI - Graphical User Interface

PLA - Polylactic Acid (3D printing filament)

ABS - Acrylonitrile Butadiene Styrene (3D printing filament)

2.0 SYSTEM SUMMARY

2.0 SYSTEM SUMMARY

The **Web-Based 3D Printing Service Ordering Management System** is a cloud-hosted platform designed to streamline 3D printing order management, automate pricing, track order progress, and facilitate direct communication between customers and staff. It replaces manual processes and Facebook Messenger interactions, allowing users to upload custom designs, choose ready-to-print models, receive cost estimates, and monitor orders in real time. The system features role-based access, guest browsing, integrated chat, email notifications, and reporting tools. Evaluated by 10 respondents including the owner, staff, a 3D artist, and invited customers, using a 5-point Likert scale, the system received high ratings across all criteria: Functionality 4.40, Reliability 4.26, Usability 4.23, Portability 4.09, and Maintainability 3.96, with an overall mean of 4.21 (Agree). These results confirm that the system effectively addresses operational challenges, providing a user-friendly, reliable, and accessible platform.

2.1 Hardware and Software Specifications

Client Devices - The system is accessible on desktop PCs, laptops, or mobile devices with a web browser and a stable internet connection. Some features may have limited display or functionality on smaller devices.

Cloud Hosting - The system is deployed on smarterasp.net to ensure online accessibility for all users.

Development Environment - The system was developed using a multi-core processor, 16 GB RAM, 250 GB SSD, Windows OS, Apache Web Server, MySQL, WordPress CMS, and programming languages including PHP, JavaScript, HTML, and CSS.

Payment Integration - Secure payment processing is enabled through PayMongo, GCash, and Maya Gateway.

2.2 User Access Levels

Admin (Owner/Administrator) - The admin has full control over the system. They can create and manage staff accounts, approve orders, verify payments, generate business analytics, and configure system settings.

Staff - Staff members handle customer orders, communicate with clients, upload gallery images, verify payments, and update the status of orders.

Registered Client - Registered clients can create an account, place 3D printing orders, upload custom designs, select from ready-made models, use the cost calculator, track order status, send and receive messages, and view transaction history.

Guest - Guests can browse the gallery, view ready-to-print models, read FAQs, and send inquiries through the contact form. They cannot place orders without registering.

2.3 Contingencies and Alternate Modes of Operation

In the event of system downtime due to website maintenance or server issues, clients will not be able to place orders or make payments until the system is restored. General internet issues on the company's side may affect internal operations, but as long as the website is online, clients can still place orders and complete payments. Staff can record and track orders manually if needed, and once the system is back online, they can update and process these orders. The system also performs daily automatic backups through the cloud hosting provider, keeping copies for 30 days to ensure data can be safely restored in case of loss or corruption.

3.0 GETTING STARTED

3.0 GETTING STARTED

This section provides step-by-step guidance on how to access and navigate the system from login to exit. It is written in clear, non-technical language for ease of use.

3.1 Logging On

1. Open a Web Browser

Recommended browsers: Google Chrome, Microsoft Edge, Mozilla Firefox, or Safari.

2. Access the Login Page

Click the Login button at the top right corner of the homepage.

3. Enter Credentials

- *Registered Clients/Staff/Admin: Enter your username and password.*
- *First-Time Users: Click Register to create a new account.*

4. Click "Login"

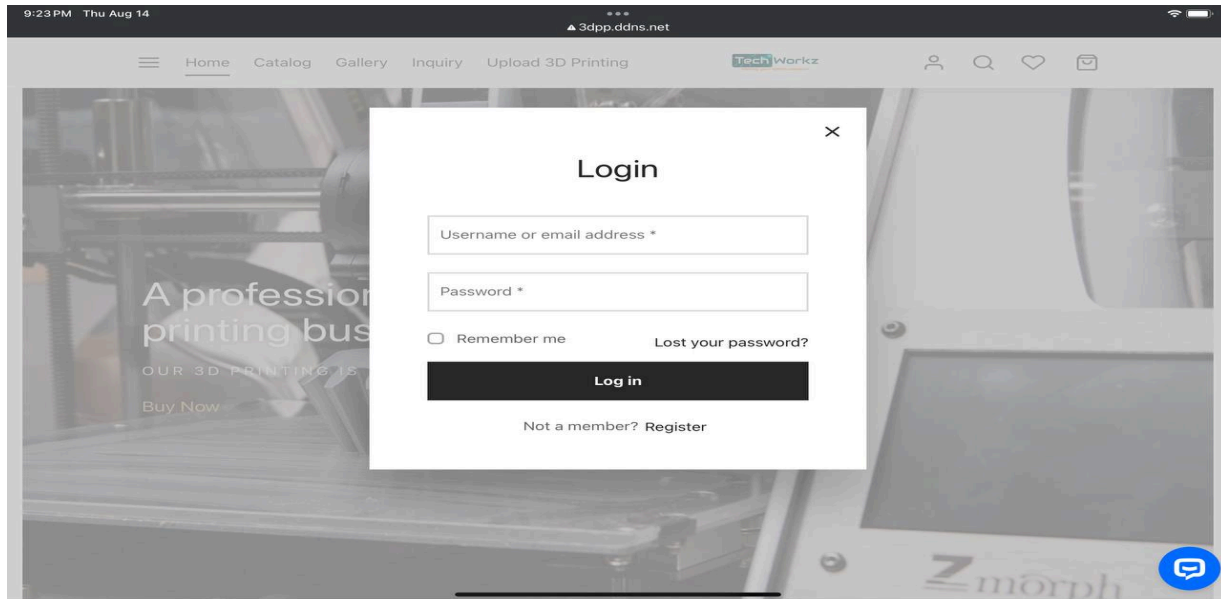
If the credentials are correct, you will be redirected to your dashboard.

5. Guest Access

Guests can browse galleries, FAQs, and ready-to-print models without logging in.

Note: Three consecutive failed login attempts will temporarily lock the account for security purposes.

Access the Login Page - Clients/Staff/Admin: Enter your username and password.



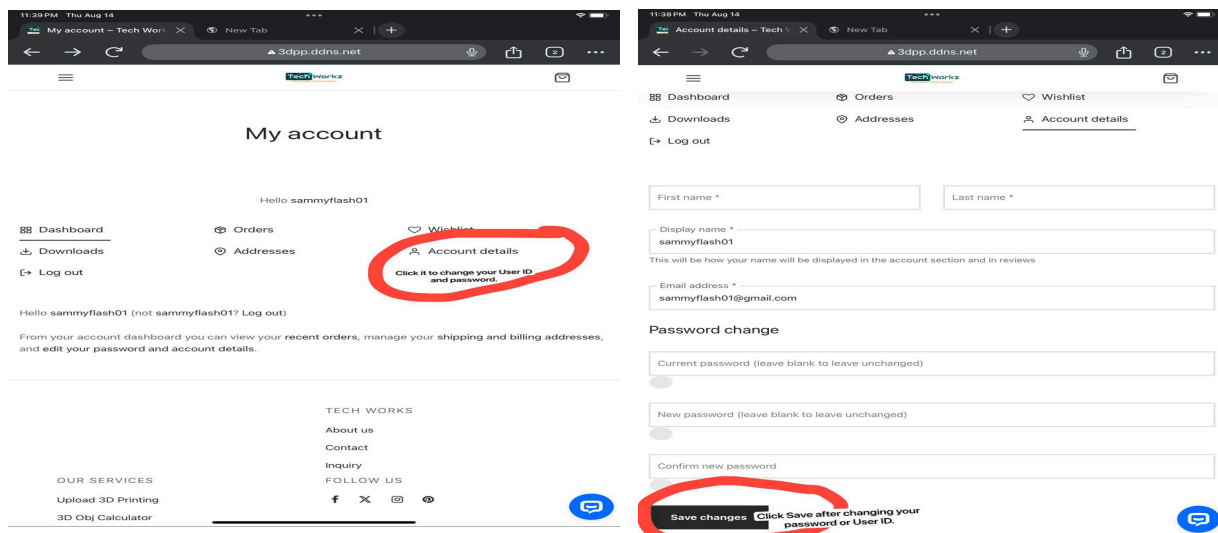
3.2 System Menu

The System Menu provides access to all the main features of the system, organized according to your user role. Registered clients can browse the gallery and ready-to-print models, upload custom orders, calculate costs, track active and completed orders, communicate with staff via messages, and update profile information. Staff and administrators can manage and update orders, verify payments, generate reports, manage user accounts, and upload project photos. The menu layout is designed for easy navigation, allowing users to quickly find and use the system's features.



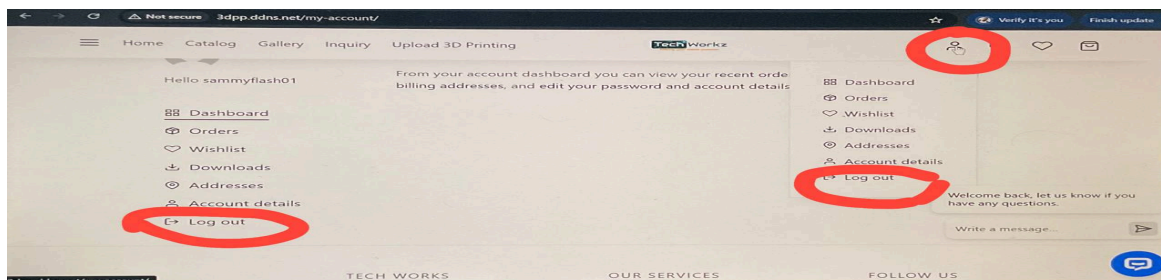
3.3 Changing User ID and Password

To change your password, first log in to your account and click **Profile** from the menu. Then select **Change Password**, enter your current password, and type your new password twice to confirm. Finally, click **Save Changes** to update your password. Passwords must be at least eight characters long and include at least one uppercase letter, one lowercase letter, and one number. For security, avoid using easily guessable passwords such as “123456” or “password.”



3.4 Exit System

To log out, click Log Out either at the top right corner or at the bottom of the dashboard. You will be redirected to the homepage. Always remember to log out, especially when using public or shared computers, to keep your account secure.



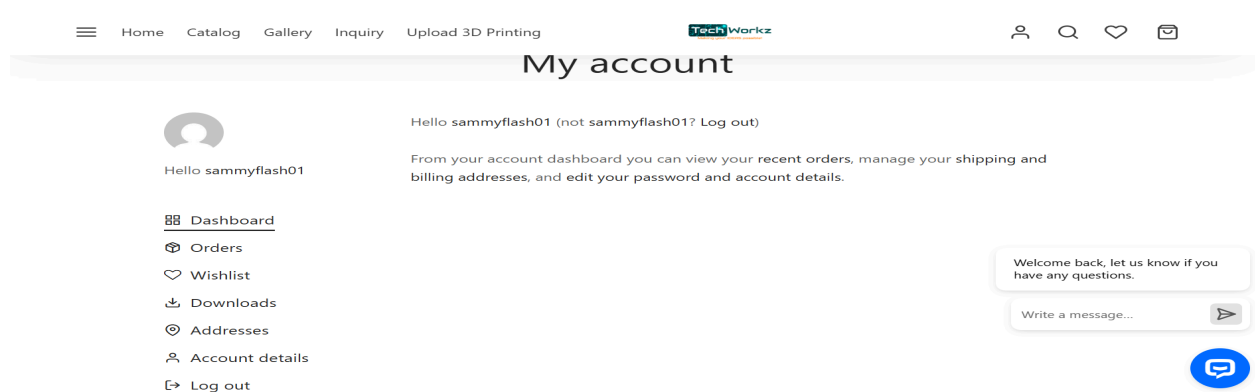
4.0 USING THE SYSTEM

This section provides detailed descriptions and instructions for all major system functions. Menu items and features vary depending on whether the user is a Client, Staff, or Admin.

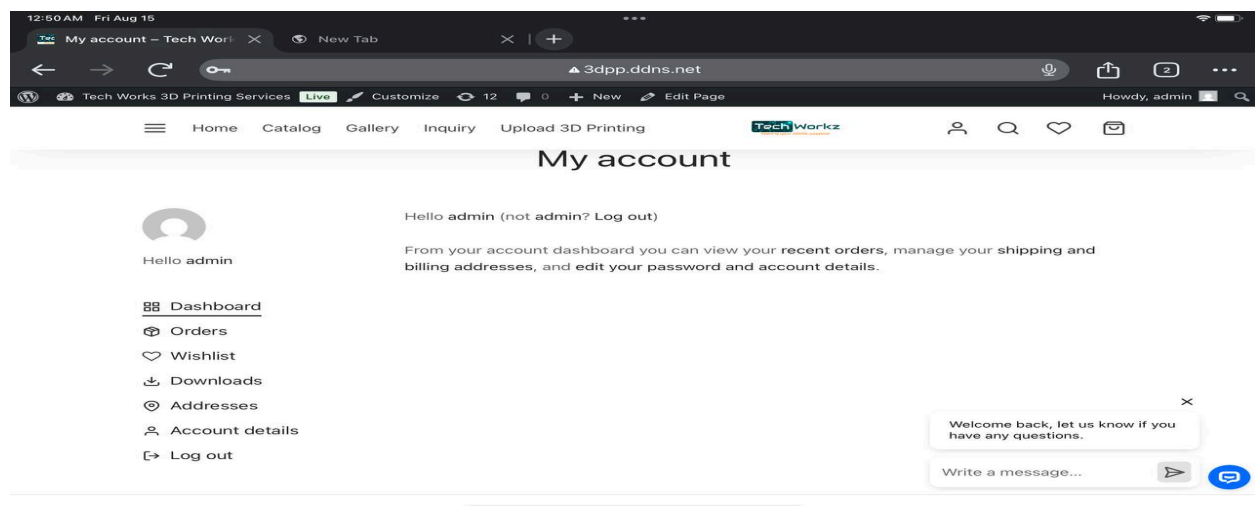
4.1 Dashboard

The Dashboard is the first page that appears after logging in and provides a quick overview of your account and recent activities. For clients, it displays active orders, pending payments, and recent messages, along with quick links to start a new order or view the gallery. For staff and administrators, it shows pending orders, recent payments, and alerts for tasks that require attention.

4.1.0 Client Dashboard

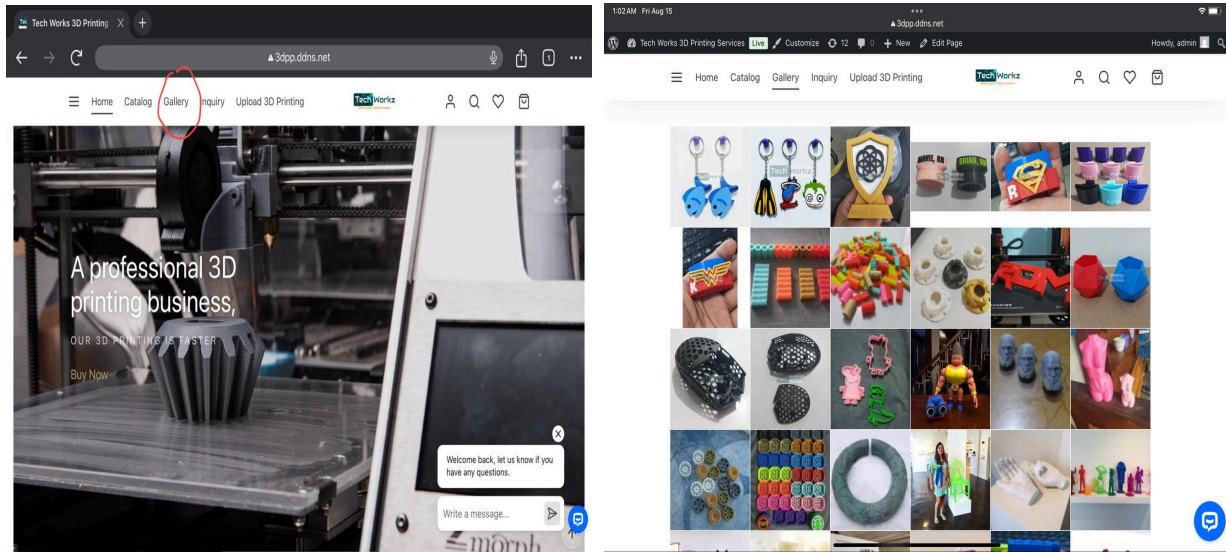


4.1.1 Staff and Administrators



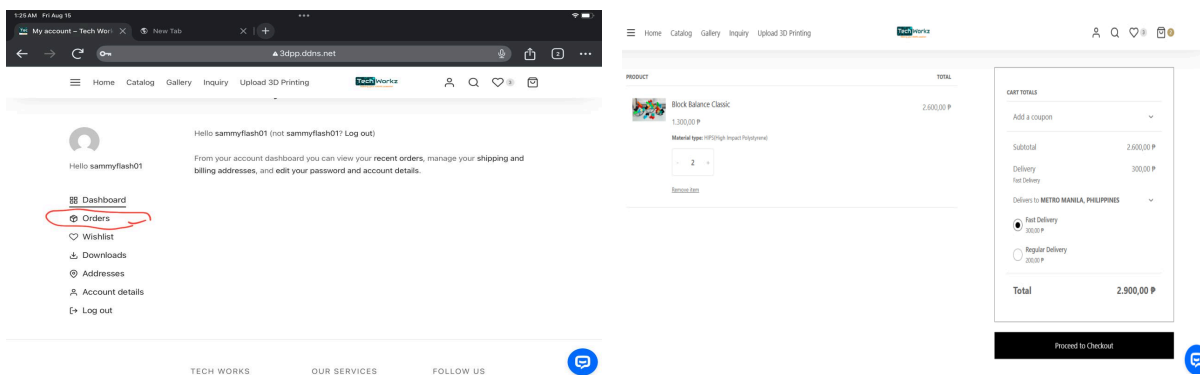
4.2 Gallery

The Gallery displays past projects and ready-to-print 3D designs. Users can access the Gallery from the system menu and browse photos or click on a project to view more details. Staff and administrators can upload new project images along with descriptions to keep the gallery updated.



4.3 Ready-to-Print Models

This section features pre-designed 3D models that clients can order instantly. To use this feature, clients can navigate to **Ready-to-Print Models** from the system menu, browse the available designs, and click **Order Now** to proceed with the order form.



4.0 Using The System

The first screenshot shows the 'Billing details' form on the left, which includes fields for first name (Sammy), last name (Flash), company name (quezon), country (Philippines), street address (SM Shopping), apartment (235), town/city (quezon City), state/country (Metro Manila), and postcode (1118). It also includes a phone number (09495214834), email address (sammyflash01@gmail.com), and a checkbox for 'Ship to a different address?'. The second screenshot shows the 'Order details' summary on the right, including the item 'Block Balance Classic - HIPS(High Impact Polystyrene)' for 2,600.00 P, shipping costs, and a total of 2,900.00 P. It also shows the billing and shipping addresses. The third screenshot shows the 'Upload 3D Printing' page with a form for Name, Email, Phone, and a message field, along with a 'Send' button.

4.4 Custom Order Submission

Clients can upload their own 3D model files for printing by selecting **Custom Order** from the menu. After uploading a supported file (STL or OBJ), they provide the required details such as dimensions, material type, and quantity, then click **Submit** to place the order.

The first screenshot shows the 'Upload 3D Printing' page with a form for Name, Email, Phone, and a message field, along with a 'Send' button. The second screenshot shows the 'Upload 3D Printing' page with a form for Name, Email, Phone, and a message field, along with a 'Send' button. The 'Send' button is highlighted with a red circle.

4.5 Cost Calculator

The Cost Calculator automatically estimates printing costs based on model details. Clients access it from the menu, enter the model's dimensions, material type, and quantity, and click **Calculate** to view the price.

Home Catalog Gallery Inquiry Upload 3D Printing

3D MODEL VIEWER
Upload 3D Model

3D PRINTING COST CALCULATOR

Cost of material

Material type: ABS (Acrylonitrile Butadiene Styrene)

Density: 1.05 g/cm³

Filament diameter: 1.75 mm

Filament length: mm

Filament weight: kg

Price: \$ / kg

Material cost: \$

Labor cost

Printing time: hrs

Cost per hour: \$ / hr

Labor cost: \$

Final price

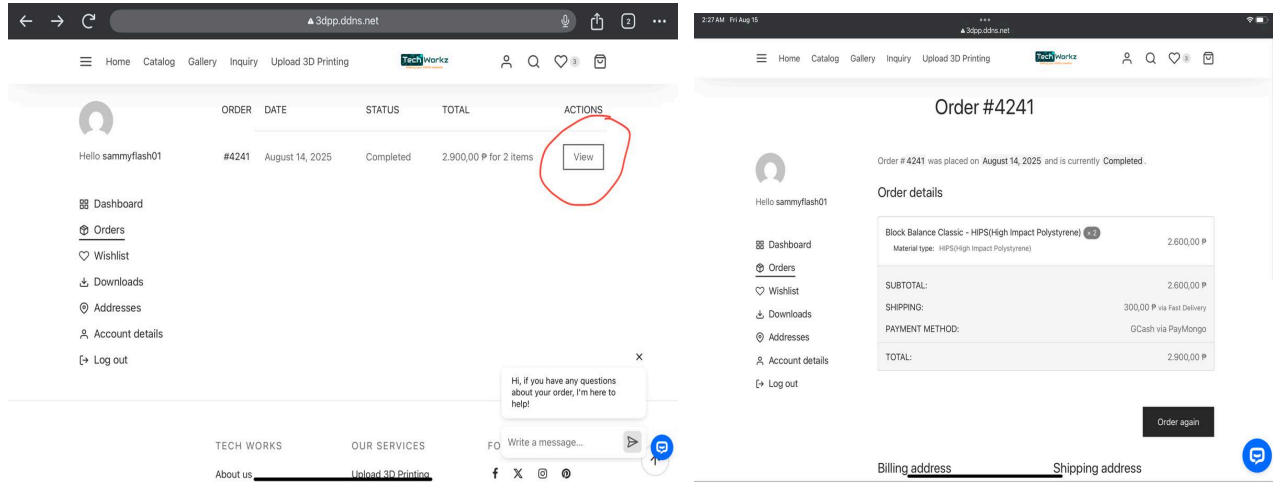
Markup: 10 %

Final price: \$

Rebuild calculator

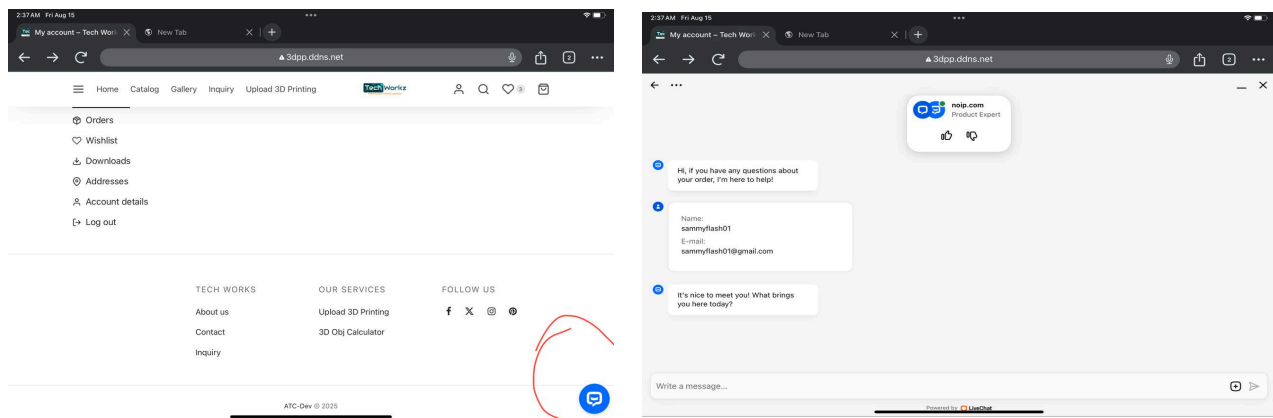
4.6 Order Tracking

Clients can monitor the progress of their orders in real-time through the **Order Tracking** section. Status updates include Pending Approval, Payment Verified, Printing in Progress, Ready for Pickup/Delivery, and Completed.



4.7 Messages/Chat

This feature allows direct communication between clients and TechWorkz staff. Clients access **Messages** from the menu, select an existing conversation or start a new one, type their message, and click **Send**. Staff and administrators can respond to messages and provide updates.



4.8 Payment & Verification

Clients make payments by selecting **Pay Now** on their order and choosing a payment method such as GCash, Maya, or Paymongo. They complete the transaction and upload proof if required. Staff and administrators verify payments by reviewing the uploaded proof and transaction reference, then clicking **Verify** to confirm the payment.

Edit Order

Activity Finish setup

Screen Options Help

Edit order Add new order

Order #4241 details

Payment via GCash via PayMongo (pay_yubGJssnHc2HkBXHYZndN). Paid on August 14, 2025 @ 5:23 pm.

Customer IP: 180.190.229.94

General

Date created:

2025-08-14 @ 17:21

Status:

Completed

Customer:

Profile - View other orders -

Sammy Flash (P55 - sammyflash01@gmail.com)

Billing

Sammy Flash

quezon

SM Shopping

235

quezon City

Metro Manila

1118

Email address:

sammyflash01@gmail.com

Phone:

09495214834

Shipping

Sammy Flash

quezon

SM Shopping

235

quezon City

Metro Manila

1118

Order attribution

Origin

Direct

Device type

Desktop

Session page views

11

Customer history

Total orders

1

Total revenue

2,900.00 ₱

Average order value

2,900.00 ₱

Item

Cost

Qty

Total

Block Balance Classic - HIPS(High Impact Polystyrene)

1,300.00 ₱

x 2

2,600.00 ₱

SKU: block-balance-classic-version-7015563

Variation ID: 3654

Material type: HIPS(High Impact Polystyrene)

Fast Delivery

300.00 ₱

Items:

Block Balance Classic - HIPS(High Impact Polystyrene) x 2

Items Subtotal:

2,600.00 ₱

Shipping:

300.00 ₱

Order Total:

2,900.00 ₱

Paid:

2,900.00 ₱

August 14, 2025 via GCash via PayMongo

4.9 Reports (Staff/Admin Only)

Staff and administrators can generate reports for sales, order history, and performance tracking. They select **Reports** from the menu, choose the report type (e.g., Daily Sales, Monthly Orders), set the date range, and click **Generate Report**.

Orders

Activity Finish setup

Date range:

Month to date (Aug 1 - 15, 2025)

vs. Previous year (Aug 1 - 15, 2024)

Show:

All orders

Orders

4

Net sales

7,900.00 ₱

Average order value

1,975.00 ₱

Average items per order

2

Orders

Month to date (Aug 1 - 15, 2025)

4

Previous year (Aug 1 - 15, 2024)

0

By day

By day

Aug 2025

Overview

Activity Finish setup

Date range:

Month to date (Aug 1 - 15, 2025)

vs. Previous year (Aug 1 - 15, 2024)

Performance

Total sales

9,100.00 ₱

Net sales

7,900.00 ₱

Orders

4

Products sold

6

Variations Sold

6

4.10 Special Instructions for Error Correction

For login issues, ensure a stable internet connection and correct credentials. File upload errors may occur if the file format or size is incorrect; only STL and OBJ formats are supported. Payment discrepancies should be reported to the TechWorkz Admin along with the transaction reference number.

4.11 Caveats and Exceptions

Orders will not start printing until payment has been verified. The system automatically logs out after 15 minutes of inactivity to ensure account security.

4.12 Input Procedures and Expected Output

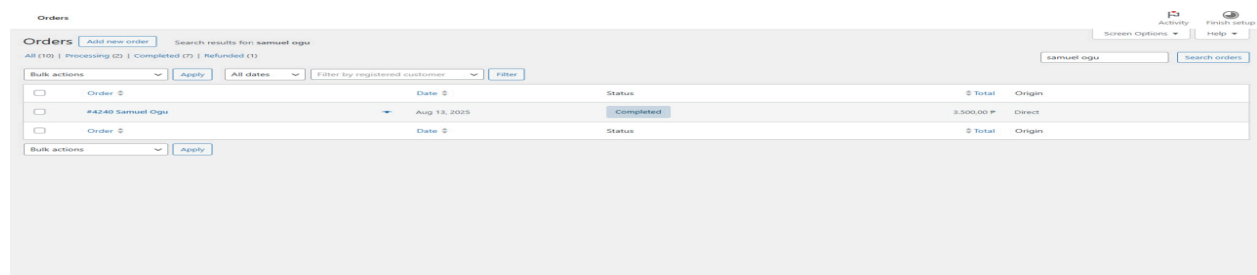
When placing an order, clients provide order details, upload files, or select a ready-made model as input. The system outputs an order confirmation page, sends an email notification, and updates the order status accordingly.

5.0 QUERYING

This section explains how staff and administrators can search for and retrieve specific information from the system. The query function allows quick access to orders, payments, and customer records.

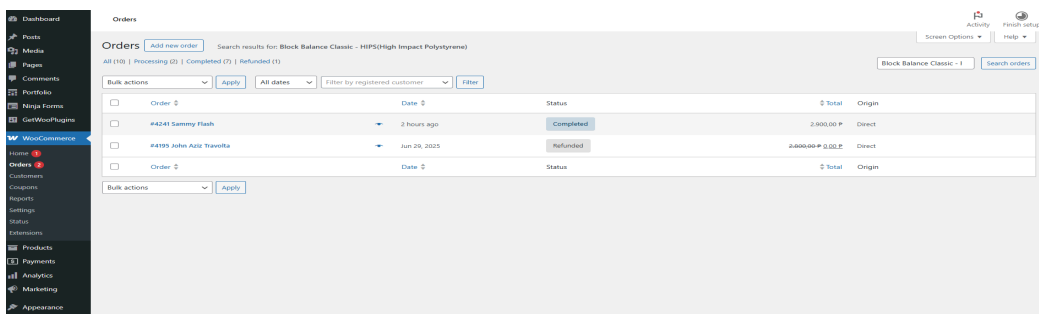
5.1 Query Capabilities

The system supports searches for orders, clients, and payments. Orders can be located by Order ID, client name, date, or status, such as “Printing in Progress” or “Completed.” Clients can be searched by name, email, or registration date. Payment records can be retrieved using the payment reference number, client name, payment method, or payment status. Additionally, when generating reports, staff and administrators can filter data by date range, order status, or material type. All searches support partial matches, so typing part of a name or ID will still return results.



5.2 Query Procedures

To search for an order, log in as staff or administrator, select **Orders** from the menu, enter the keyword in the search bar at the top right of the page, and press Enter or click the search icon. Matching results will appear in a table showing Order ID, client name, status, and date.



6.0 REPORTING

The reporting feature allows staff and administrators to generate various reports for monitoring business performance, tracking orders, and managing finances. Reports can be filtered by date range, order status, and material type to focus on relevant information.

6.1 Types of Reports Available

The system provides several types of reports. The **Daily Sales Report** lists all completed orders for the day, including order IDs, client names, payment amounts, and payment methods. The **Monthly Sales Report** summarizes sales for the month, showing total revenue, completed orders, and best-selling materials. The **Order History Report** displays all orders within a selected time frame, including both completed and pending transactions. The **Customer Activity Report** shows customer engagement, including the number of orders, purchase frequency, and order value. Optionally, the **Material Usage Report** tracks the quantities of materials such as PLA and ABS used during the selected period.

6.2 Generating a Report

To generate a report, log in as staff or administrator and select **Reports** from the menu. Choose the report type, set the start and end dates, apply additional filters if needed (such as order status or material type), and click **Generate Report**.

